CHANDRASHEKAR BALA

Cybersecurity Researcher | Offensive Security | VAPT & AI Security

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SUMMARY

Cybersecurity professional with hands-on experience in offensive security, penetration testing, vulnerability assessment, web application security, network security, exploit research, privilege escalation, post-exploitation, threat research, digital forensics, and security engineering. Experienced in reconnaissance, service enumeration, vulnerability validation, controlled exploitation, attack-path analysis, web application testing, proof-of-concept development, OSINT investigations, network traffic analysis, and technical security reporting. Currently serving as a Cybersecurity Intern at Networkwalks Technologies, contributing to network security, vulnerability assessment, penetration testing, security operations, incident response, digital forensics, and risk assessment activities. Strong background in Linux, Windows, Python, Bash, C, Git, Nmap, Wireshark, Burp Suite, OWASP ZAP, Metasploit, GCC, and GDB.

EXPERIENCE

Cybersecurity Intern — Networkwalks Technologies Sep'26 – Present

Cybersecurity Internship | VAPT, Network Security, Security Operations & Incident Response

- Work across network security, vulnerability assessment, penetration testing, security operations, incident response, digital forensics, and risk assessment activities.
- Perform security assessment, reconnaissance, vulnerability validation, technical analysis, investigation, and security documentation across assigned cybersecurity work.
- Apply offensive-security techniques to identify weaknesses, validate findings, analyze attack paths, assess impact, and document remediation considerations.

Cybersecurity Researcher — Independent Jul'24 – Present

Independent Cybersecurity Research & Security Engineering | Self-employed

- Conduct hands-on cybersecurity research across offensive security, vulnerability assessment, penetration testing, web application security, network security, wireless security, system security, and security engineering.
- Perform reconnaissance, service enumeration, vulnerability assessment, vulnerability validation, controlled exploitation, privilege escalation, post-exploitation, and attack-path analysis.
- Conduct web application security testing with Burp Suite and OWASP ZAP across SQL injection, XSS, authentication, authorization, and access-control scenarios.
- Analyze HTTP requests, responses, parameters, sessions, authentication flows, and application behavior to identify security weaknesses, exploitation conditions, and attack paths.
- Perform OSINT and threat research involving domains, IP addresses, infrastructure, threat reports, malware, attacker TTPs, and security indicators.
- Analyze packet captures and network traffic using Wireshark to investigate protocols, communication patterns, suspicious connections, and attacker behavior.
- Perform Windows forensic and malware-analysis exercises involving registry artifacts, file systems, disks, suspicious files, and static/dynamic analysis.
- Develop Python and Bash tooling for reconnaissance, security testing, repeatable analysis, proof-of-concept development, and security research.
- Document findings, technical evidence, exploitation conditions, attack paths, security impact, remediation recommendations, and testing methodology.

Security Researcher — Independent Jul'26 – Aug'26

Independent Security Research & Security Engineering | Self-employed

- Completed Linux kernel compatibility and wireless-security engineering involving a Realtek RTL8812BU/RTL8822BU USB Wi-Fi driver.
- Analyzed driver source and implemented targeted compatibility fixes across 13 source files for newer Linux kernel APIs and build requirements.
- Diagnosed compiler, kernel API, and module-build issues using C, Linux kernel headers, GCC, conditional compilation, Git, and Linux kernel build workflows.
- Built and validated the modified kernel module on Kali Linux using real RTL8812BU hardware, including monitor mode, packet injection, packet capture, Wireshark, and Aircrack-ng workflows.

PROJECTS

Linux Kernel Compatibility & Wireless Security Engineering Jul'26 – Aug'26

C | Linux Kernel | RTL8812BU/RTL8822BU | Kali Linux | Git

- Modified an existing Realtek USB Wi-Fi driver to address compatibility issues introduced by newer Linux kernel APIs and build requirements.
- Analyzed kernel-facing C code and implemented targeted compatibility changes across 13 source files while preserving the existing driver architecture and functionality.
- Built, installed, and validated the modified kernel module on Kali Linux using real RTL8812BU hardware.
- Validated monitor mode, packet injection, packet capture, Wireshark analysis, and Aircrack-ng workflows for wireless-security testing.

Web Application Security Assessment Lab Feb'26 – Present

PortSwigger Web Security Academy | Burp Suite | OWASP ZAP

- Conduct hands-on web application security testing covering SQL injection, XSS, authentication, authorization, access-control, and OWASP Top 10 attack scenarios.
- Analyze HTTP requests, responses, parameters, sessions, authentication flows, and application behavior to identify security weaknesses and attack paths.
- Reproduce findings, capture technical evidence, assess security impact, and document exploitation conditions and remediation considerations.

PROJECTS & TECHNICAL RESEARCH

Vulnerability Assessment & Privilege Escalation Lab Dec'25 – Present
VulnHub Virtual Machines | Nmap | Linux | Metasploit

- Perform reconnaissance, service enumeration, vulnerability assessment, system enumeration, and controlled exploitation across isolated virtual machines.
- Perform Linux privilege escalation and post-exploitation analysis and trace attack paths from initial access through elevated privileges.
- Validate vulnerabilities and document findings, exploitation paths, security impact, evidence, and hardening recommendations.

Buffer Overflow Exploit Development Completed
C | Linux | GDB | Exploit Development

- Analyze stack-based buffer-overflow vulnerabilities and vulnerable program behavior in Linux environments.
- Use GDB and memory-analysis techniques to examine stack layout, control flow, registers, and exploitation conditions and develop proof-of-concept exploitation workflows.

Enterprise Network Penetration Testing Lab Completed
Nmap | Metasploit | Linux | Network Security

- Conduct reconnaissance, service enumeration, vulnerability assessment, and attack-path analysis against a controlled enterprise-style network.
- Investigate exposed services and security weaknesses, validate findings through controlled testing, and document attack paths and remediation.
- Use Nmap and Metasploit to investigate vulnerable services and analyze exploitation paths within an isolated environment.

Fake News Detection System — Capstone Project Apr'24
Machine Learning | Python | Natural Language Processing

- Designed and implemented a machine-learning system to classify news articles as legitimate or fake using labeled datasets.
- Performed text preprocessing, feature extraction, model training, and evaluation using Python.
- Analyzed classification results and model performance to identify misclassifications and improve the reliability of the prediction workflow.

SKILLS

Offensive Security	Reconnaissance, Service Enumeration, Vulnerability Assessment, Penetration Testing, Security Testing, Vulnerability Validation, Controlled Exploitation, Privilege Escalation, Post-Exploitation, Attack-Path Analysis
Web & API Security	OWASP Top 10, SQL Injection, Cross-Site Scripting, Authentication, Authorization, Access Control, HTTP/HTTPS, Request/Response Analysis, Burp Suite, OWASP ZAP
Network & Wireless Security	TCP/IP, DNS, DHCP, Ports and Protocols, Routing, Switching, Subnetting, CIDR, Network Reconnaissance, Network Traffic Analysis, Packet Capture, Monitor Mode, Packet Injection
Exploit Development	Buffer Overflow Analysis, Memory Analysis, GDB, C, Linux, Proof-of-Concept Development, Exploitation Research
Threat & Adversary Research	Threat Intelligence, OSINT, Threat Report Analysis, TTP Analysis, Infrastructure Analysis, Adversary Research, MITRE ATT&CK
Forensics & Malware	Windows Forensics, Registry Analysis, File-System Analysis, Disk Analysis, Malware Analysis, Static/Dynamic Analysis
Programming & Systems	Python, Bash, C, SQL, Linux, Windows, Git, GCC, GDB, Linux Kernel Headers, Kernel Build Workflows
Security Tooling	Nmap, Wireshark, Burp Suite, OWASP ZAP, Metasploit, Nessus, Aircrack-ng, GDB
AI / ML & Frameworks	Machine Learning, NLP, Text Preprocessing, Feature Extraction, Classification, NIST Cybersecurity Framework, MITRE ATT&CK

TOOLS & TECHNICAL STACK

Web / Application	Burp Suite, OWASP ZAP, HTTP/HTTPS analysis, request/response analysis, authentication and access-control testing
Network / Discovery	Nmap, Wireshark, packet capture, service enumeration, network reconnaissance, protocol analysis
Exploitation	Metasploit, GDB, controlled exploitation, privilege escalation, post-exploitation, attack-path analysis
Assessment / Validation	Nessus, vulnerability assessment, vulnerability validation, security testing, evidence collection, technical reporting
Threat Research	MITRE ATT&CK, OSINT, TTP analysis, threat-report analysis, infrastructure analysis, adversary research
Programming / Systems	Python, Bash, C, SQL, Linux, Kali Linux, Windows, GCC, GDB, Git, Linux kernel build workflows

ADDITIONAL TRAINING

- **Java Full Stack Developer Training** — Feb 2023 | CipherSchools — Java-based full-stack development and databases.
- **Google Cloud Facilitator Program** — May 2022 | Google — cloud computing and infrastructure.

CORE SECURITY CAPABILITIES

VAPT Delivery	Reconnaissance, service enumeration, vulnerability assessment, vulnerability validation, controlled exploitation, privilege escalation, post-exploitation, attack-path analysis, technical reporting
Web Security Testing	HTTP analysis, SQL injection, XSS, authentication, authorization, access-control testing, Burp Suite, OWASP ZAP
Network Testing	Network reconnaissance, service enumeration, exposed-service analysis, packet analysis, Nmap, Wireshark, Metasploit
Exploit Research	Buffer-overflow analysis, memory analysis, GDB, C, proof-of-concept development, exploitation research
Threat Research	Threat intelligence, OSINT, TTP analysis, adversary behavior, infrastructure analysis, MITRE ATT&CK
Security Engineering	Linux security, wireless security, kernel compatibility, C systems programming, Python/Bash automation, technical troubleshooting
Security Reporting	Technical evidence, vulnerability impact, exploitation conditions, attack paths, remediation recommendations, security documentation

CERTIFICATIONS & PROFESSIONAL TRAINING

Tata — Cybersecurity Analyst Job Simulation Aug 2026
Forage

Practical simulation covering IAM strategy assessment, solution design, implementation planning, and secure access workflows.

Google Cybersecurity Professional Certificate Feb 2026
Coursera

Security, networking, Linux, SQL, Python, SIEM workflows, threat detection, incident response, and SOC workflows.
Key Learning: Log analysis, alert triage, vulnerability management, risk assessment, access control, and NIST Cybersecurity Framework.

Introduction to Blockchain and Web3 May 2024
edX

Blockchain, decentralized systems, Web3 architecture, and security considerations.

Natural Language Processing Specialization Apr 2023
Coursera

Text preprocessing, classification, and sequence models using Python.

Java Full Stack Developer Training Feb 2023
CipherSchools

Java-based full-stack development and databases.

Google Cloud Facilitator Program May 2022
Google

Cloud computing and infrastructure.

SELECTED TECHNICAL EVIDENCE

- **Linux kernel engineering:** analyzed kernel-facing C code, diagnosed API/build compatibility issues, implemented targeted fixes across 13 source files, and validated the resulting module on real RTL8812BU hardware.
- **Wireless security validation:** validated monitor mode, packet injection, packet capture, Wireshark traffic analysis, and Aircrack-ng workflows using physical RTL8812BU hardware on Kali Linux.
- **Web security testing:** used Burp Suite and OWASP ZAP to analyze HTTP requests/responses and test SQL injection, XSS, authentication, authorization, and access-control vulnerabilities.
- **Vulnerability assessment:** performed reconnaissance, service enumeration, vulnerability validation, Linux privilege escalation, post-exploitation analysis, and attack-path documentation across isolated virtual machines.
- **Exploit research:** used C, Linux, GDB, stack/memory analysis, and controlled proof-of-concept workflows to study buffer-overflow exploitation conditions.
- **Threat research:** conducted OSINT and threat-report analysis, mapped TTPs to MITRE ATT&CK, and investigated attacker behavior and security indicators.
- **Security automation:** developed Python/Bash scripts for reconnaissance, repeatable testing, proof-of-concept development, and security research workflows.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science and Engineering 2024
Relevant Coursework: Network Security, Ethical Hacking, Penetration Testing, Machine Learning, Automation — Jalandhar, Punjab

Intermediate (MPC), Telangana Board of Intermediate Education 2020
Mathematics, Physics, Chemistry — Warangal, Telangana

TECHNICAL FOCUS

Offensive Security	Reconnaissance, service enumeration, vulnerability assessment, vulnerability validation, controlled exploitation, privilege escalation, post-exploitation
Web Application Security	OWASP Top 10, HTTP analysis, SQL injection, XSS, authentication, authorization, access control, Burp Suite, OWASP ZAP
Network Security	Network reconnaissance, service enumeration, packet capture, traffic analysis, Nmap, Wireshark, Metasploit
Exploit Research	Buffer-overflow analysis, GDB, C, memory analysis, proof-of-concept development, exploitation research
Security Engineering	Linux, wireless security, kernel compatibility, Python/Bash automation, C, Git, GCC
AI Security	AI/ML security research direction, NLP, adversarial testing, LLM security research, prompt-injection and RAG security research